GenMol QED Bayesian Shrinkage Ablation

Validated 50,000-step-checkpoint comparison of the released update, online running mean, and four fixed Bayesian shrinkage strengths.

Protocol field	Locked value
Oracle and budget	QED; 1,000 unique canonical molecules per variant-seed run
Arms	released, running_mean, shrink1, shrink3, shrink10, shrink30
Paired seeds	0, 1, 2 (18 completed runs)
Bayesian prior	fixed proxy mean 0.5; lambda in {1, 3, 10, 30}
Population and warmup	population=100; warmup=100; legacy off-by-one=True
Sampling controls	temperature=1.2; randomness=2.0; guidance=2.0; gamma=0.0
Molecule bounds	10 to 30 under the runner convention
Loop and persistence	max iterations=10000; report/checkpoint every 100/100 calls; durable events=True
Fragment statistics	minimum support=1; legacy seed count=1; parent control=False
Sampling and duplicates	canonical fragment string before uniform sampling; statistical: one update per unique canonical child; released: repeat cached-child decomposition, matching release
Primary endpoint	normalized trapezoidal AUC of mean top-10 QED vs oracle calls
Scientific status	Exploratory three-seed QED Bayesian prior-strength sensitivity study using the final 50k checkpoint, a 1,000-unique-oracle-call budget, a 100-iteration warmup, legacy seed count 1, and a fixed neutral external proxy prior mean of 0.5 because the original ZINC molecule-level score distribution is unavailable. It is not paper-comparable and is not confirmatory.

Aggregate scalar metrics

Normalized AUC metrics

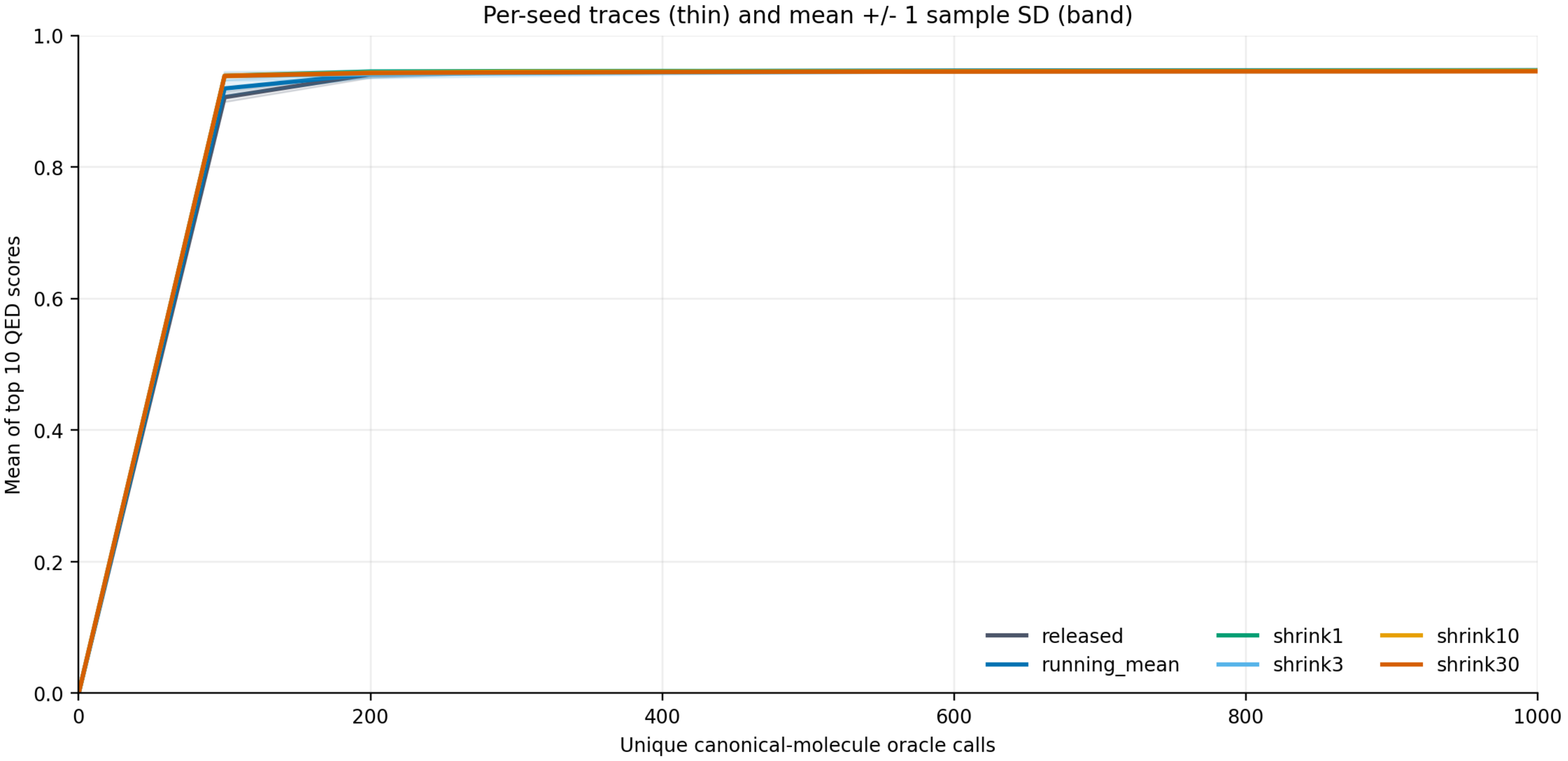
Variant	lambda	Top-1 AUC	Top-10 AUC	Top-100 AUC	Paired top-10 AUC delta vs mean
released	n/a	0.899727 +/- 0.000983	0.894028 +/- 0.001708	0.853075 +/- 0.003966	-0.001842 +/- 0.003984
running_mean	0	0.899855 +/- 0.001424	0.895869 +/- 0.002388	0.857370 +/- 0.004860	0.000000 +/- 0.000000
shrink1	1	0.900677 +/- 0.000216	0.898073 +/- 0.000738	0.854082 +/- 0.005275	0.002204 +/- 0.002767
shrink3	3	0.900396 +/- 0.000473	0.896511 +/- 0.002984	0.850591 +/- 0.006188	0.000642 +/- 0.001985
shrink10	10	0.900686 +/- 0.000136	0.897395 +/- 0.000801	0.848850 +/- 0.002930	0.001526 +/- 0.001807
shrink30	30	0.900184 +/- 0.000098	0.896810 +/- 0.001299	0.851198 +/- 0.002153	0.000941 +/- 0.001751

Final-budget top-k metrics

Variant	lambda	Final top-1	Final top-10	Final top-100	Paired final top-10 delta vs mean
released	n/a	0.948252 +/- 0.000240	0.946821 +/- 0.000549	0.933910 +/- 0.003292	-0.000376 +/- 0.000928
running_mean	0	0.948149 +/- 0.000316	0.947197 +/- 0.000532	0.938896 +/- 0.001739	0.000000 +/- 0.000000
shrink1	1	0.948085 +/- 0.000235	0.946953 +/- 0.000143	0.925878 +/- 0.001675	-0.000243 +/- 0.000389
shrink3	3	0.947811 +/- 0.000491	0.945883 +/- 0.001473	0.923429 +/- 0.006982	-0.001314 +/- 0.001042
shrink10	10	0.948131 +/- 0.000187	0.946290 +/- 0.000463	0.922418 +/- 0.001726	-0.000907 +/- 0.000382
shrink30	30	0.947562 +/- 0.000103	0.945458 +/- 0.000724	0.922835 +/- 0.002996	-0.001739 +/- 0.000508

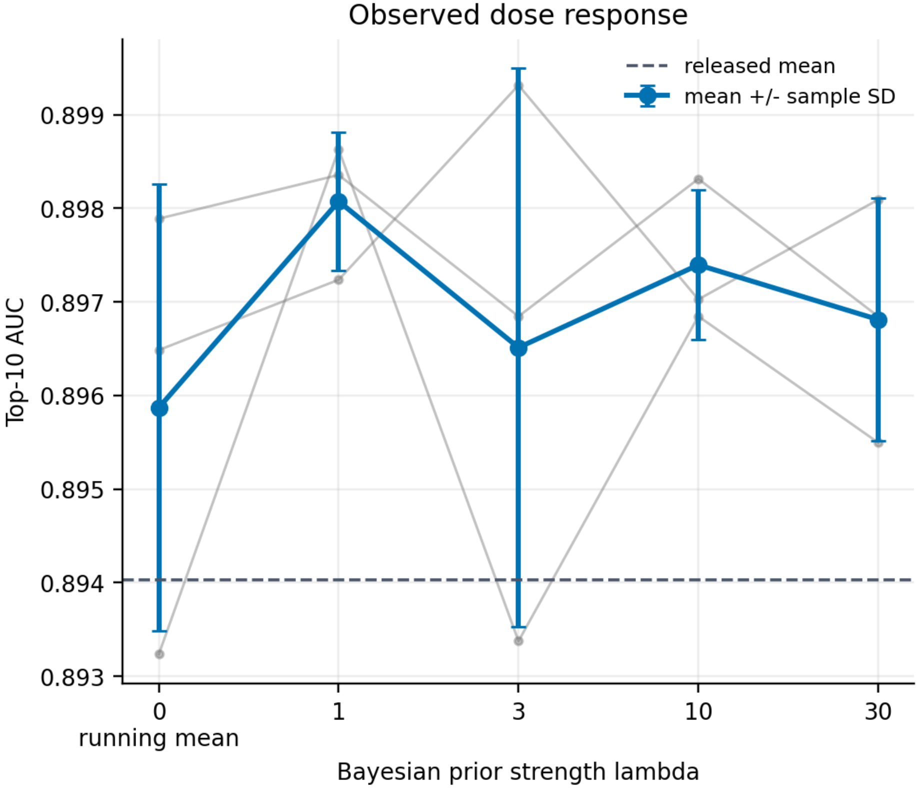
Values are mean +/- sample SD across three paired seeds. The sample size is intentionally too small for a confirmatory significance claim.

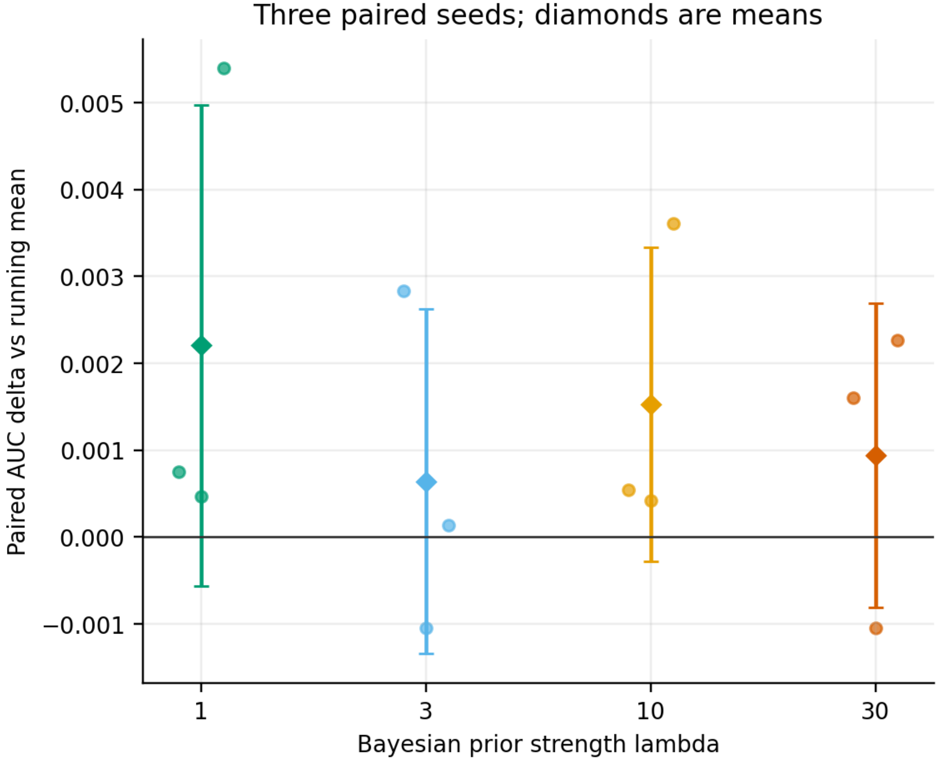
Top-10 learning curves



The origin is fixed at zero and AUC is normalized by the common 1,000-call budget, matching the experiment-I/O convention. All arms spend the same unique-molecule budget; no parent-scoring arms are included.

Bayesian shrinkage dose response





Variant	lambda	Paired AUC delta	Paired final top-10 delta	AUC deltas by seed
released	n/a	-0.001842 +/- 0.003984	-0.000376 +/- 0.000928	s0=-0.001969, s1=-0.005760, s2=+0.002204
running_mean	0	0.000000 +/- 0.000000	0.000000 +/- 0.000000	s0=+0.000000, s1=+0.000000, s2=+0.000000
shrink1	1	0.002204 +/- 0.002767	-0.000243 +/- 0.000389	s0=+0.000751, s1=+0.000467, s2=+0.005395
shrink3	3	0.000642 +/- 0.001985	-0.001314 +/- 0.001042	s0=+0.002830, s1=-0.001044, s2=+0.000140
shrink10	10	0.001526 +/- 0.001807	-0.000907 +/- 0.000382	s0=+0.000541, s1=+0.000425, s2=+0.003611
shrink30	30	0.000941 +/- 0.001751	-0.001739 +/- 0.000508	s0=+0.001607, s1=-0.001045, s2=+0.002261

Per-seed results

Variant	Seed	lambda	Top-10 AUC	Top-1	Top-10	Top-100	GPU shared	Run ID
released	0	n/a	0.894516	0.948331	0.947121	0.933512	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:released:seed0
released	1	n/a	0.892129	0.947982	0.946188	0.930835	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:released:seed1
released	2	n/a	0.895438	0.948442	0.947155	0.937383	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:released:seed2
running_mean	0	0	0.896485	0.948331	0.947534	0.940473	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:running_mean:seed0
running_mean	1	0	0.897889	0.948331	0.947472	0.939184	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:running_mean:seed1
running_mean	2	0	0.893234	0.947785	0.946584	0.937031	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:running_mean:seed2
shrink1	0	1	0.897236	0.948347	0.947047	0.924519	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink1:seed0
shrink1	1	1	0.898356	0.947890	0.947024	0.925364	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink1:seed1
shrink1	2	1	0.898629	0.948020	0.946789	0.927750	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink1:seed2
shrink3	0	3	0.899314	0.948331	0.947393	0.931419	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink3:seed0
shrink3	1	3	0.896844	0.947745	0.945804	0.918496	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink3:seed1
shrink3	2	3	0.893374	0.947357	0.944451	0.920374	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink3:seed2
shrink10	0	10	0.897025	0.948100	0.946186	0.921274	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink10:seed0
shrink10	1	10	0.898313	0.947962	0.946796	0.924403	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink10:seed1
shrink10	2	10	0.896845	0.948331	0.945887	0.921575	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink10:seed2
shrink30	0	30	0.898092	0.947682	0.946276	0.926255	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink30:seed0
shrink30	1	30	0.896844	0.947506	0.945202	0.921575	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink30:seed1
shrink30	2	30	0.895495	0.947500	0.944896	0.920675	True	fragment_vocab_qed_50k_bayes_strength_1k_v1:qed:shrink30:seed2

Every scalar above was reconciled across the immutable CSV, collection manifest, and the collector-validated summary. Individual seeds remain visible so stability is not hidden by an average.

Integrity and limitations

Provenance field	Validated value
Collection manifest	/home/aidar.alimbayev/Documents/genmolv2/experiments/fragment_vocabulary/results/qed_50k_bayes_strength_1k_v1/collection_manifest.json | sha256 c4993303a716641a1b17c8d17d2d65b72d823f8f52ae7db5197ee97628dba047
Immutable results CSV	results.969e99cced94c31bf868a79bc0c36f9daf7b3db52454517350bdb6fbb7bfd2a6.csv | sha256 969e99cced94c31bf868a79bc0c36f9daf7b3db52454517350bdb6fbb7bfd2a6
Matrix	/home/aidar.alimbayev/Documents/genmolv2/run_sources/qed_50k_bayes_47f7692/experiments/fragment_vocabulary/configs/qed_50k_bayes_strength_1k_v1.yaml | sha256 26f3ad9a9d21ee203bd4ba23b110851c67263b6d20e5af5d03660e3c592aa59d
Model	/home/aidar.alimbayev/Documents/genmolv2/outputs/paper_v1/checkpoints/50000.ckpt | sha256 8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6
Vocabulary	/home/aidar.alimbayev/Documents/genmolv2/run_sources/qed_50k_bayes_47f7692/scripts/exps/pmo/vocab/qed.csv | sha256 6b98420f3fa88835e50cb28b960225dc6c7857a29f3057fb1ca22b4211c1806c
Code identity	commit 47f7692cc21eda826511ce311c9dfa7573df4c0c | dirty=False | tracked diff e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
Trajectory evidence	18 summary files re-hashed; endpoints and normalized trapezoidal AUCs recomputed
Physical GPU mapping	3=GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959; 4=GPU-d32809ef-c445-c02a-9ec0-9ac9cf201181; 5=GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c; 6=GPU-d7e40ce5-9260-a208-16a0-3372e3f204de
Pre-launch gate	18/18 launches at 4-9% utilization (<10%) with at least 47163 MiB free; 18/18 shared with pre-existing low-utilization processes
Runtime record	controller span 1070.0 s; sum of per-run elapsed times 3444.0 s; median 174.9 s (range 147.2-299.1); resumes=0. Wall-time comparisons are suppressed because GPUs were shared.

Paper reference point

GenMol Table 13 reports QED PMO AUC top-10 of 0.942 +/- 0.000 across three runs at a 10,000-oracle-call budget. That value is included only as context: this report uses 1,000 calls, a local 50,000-step checkpoint, and a different fragment-vocabulary update study, so a numeric gap is not an efficacy comparison.

Scientific caveats

- This is an exploratory QED-only ablation (one of 23 PMO tasks), not a paper-scale PMO reproduction.
- Each arm has only three paired seeds, and each variant-seed run has 1,000 unique oracle calls; no p-values or confirmatory efficacy claims are reported.
- The Bayesian arms use a fixed neutral proxy prior mean of 0.5 because the original ZINC molecule-level prior is unavailable.
- Legacy seed fragments use the declared count-one approximation; this is not exact continuation of paper Equation 5 statistics.
- The locally trained 50,000-step checkpoint differs from the paper's complete hardware/training protocol.
- Whole-molecule scores associated with fragments are adaptive and non-causal; this report measures optimization outcomes, not fragment causality.
- At least one launch shared a GPU; score metrics remain usable, but wall-time comparisons are suppressed.

Interpretation boundary: observed differences describe this locked exploratory matrix only. They neither reproduce the paper's 23-task, 10,000-call PMO aggregate nor establish that Bayesian shrinkage is generally superior.